

Basic Principles of Medicine1

Module: Cardiovascular, Pulmonary, Renal

DLA16

Pulmonary Circulation

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Learning Objectives

By the end of this DLA students should be able to:

SOM.MK.II.BPM1.3.CPR.2.PHYS.0213	Contrast the systemic and pulmonary circulations with respect to pressures, resistance to blood flow, and response to hypoxia.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0214	1.Explain the regional differences in pulmonary blood flow in an upright person. 2.Define zones I, II, and III in the lung, with respect to pulmonary vascular pressure and alveolar pressure.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0215	Explain how distention and recruitment of pulmonary vessels influences pulmonary vascular resistance with alterations in cardiac output or pulmonary arterial pressure. Explain in terms of distention and recruitment of pulmonary vessels. Identify the zones in which these two mechanisms apply.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0216	Explain how pulmonary vascular resistance in alveolar and extra-alveolar blood vessels changes with lung volume.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0217	Describe the consequence of hypoxic pulmonary vasoconstriction on the distribution of pulmonary blood flow.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0218	Explain the development of pulmonary edema by a) increased hydrostatic pressure, b) increased permeability, c) impaired lymphatic outflow or increased central venous pressure, and d) hemodilution (e.g., with saline volume resuscitation).
SOM.MK.II.BPM1.3.CPR.2.PHYS.0219	Define right-to-left shunts, anatomic and physiological shunts, and physiologic dead space (wasted ventilation).
SOM.MK.II.BPM1.3.CPR.2.PHYS.0365	Describe the consequences of each of the above in pulmonary gas exchange.



Recommended Reading

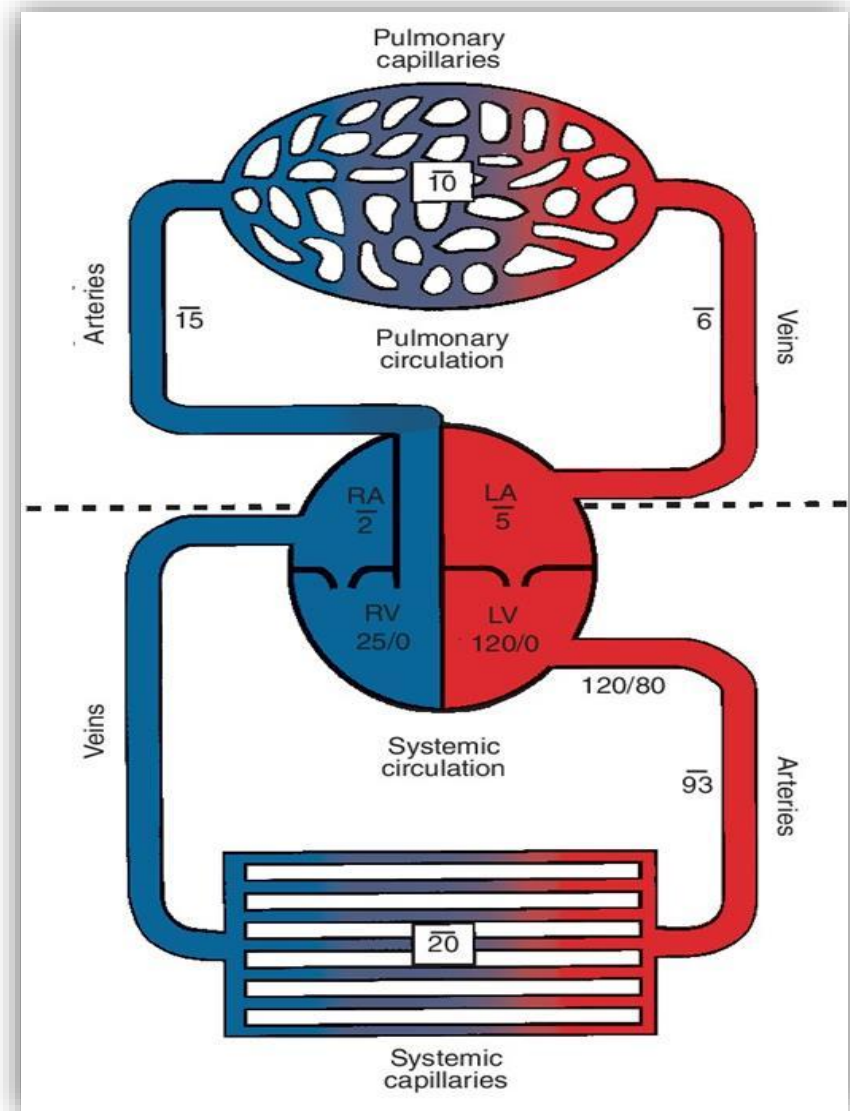
Rhoades and Bell, *Medical Physiology: Principles for Clinical Medicine*, Sixth Edition, © Lippincott, Williams & Wilkins (2023), Chapters 20

- <https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253088770&bookid=3211>



Pulmonary Circulation

- Lungs receive 100% of cardiac output via the pulmonary circulation
- The pulmonary circulation has some unique features. Unlike the systemic circulation, the pulmonary circulation is a low-pressure and low-resistance system.



Hemodynamic features of the pulmonary circulation

Pulmonary circulation is characterized as normally dilated, whereas the systemic circulation is characterized as normally constricted. The units for pressure are given in mm Hg, and a bar over the number indicates mean pressure. LA, left atrium; LV, left ventricle; RA, right atrium; RV, right ventricle.

Pulmonary vs. Systemic Circulation -Pressure

- In the pulmonary circulation blood moves from the pulmonary artery to the left atrium
- In the systemic circulation blood moves from the aorta to the right atrium

Location	Mean Pressure (mm Hg)
Pulmonary Artery	15
Capillary Start	12
Capillary End	9
Left Atrium	8

Driving Pressure: $15-8=7$ mm Hg

Location	Mean Pressure (mm Hg)
Aorta	95
Capillary Start	35
Capillary End	15
Right Atrium	2

Driving Pressure : $95-2=93$ mm Hg

- Blood moves along a pressure gradient within the circulatory system. Pulmonary circulation is a low-pressure system while the systemic pressure operates with high pressure.



Pulmonary vs. Systemic Circulation -Resistance

Resistance = Pressure gradient / Flow

$$R = \Delta P / \dot{Q}$$

- Flow (cardiac output) = 5L/min or 83 ml/s

➡ Systemic Resistance = 93 mm Hg/83 ml/s = 1.1 peripheral resistance unit

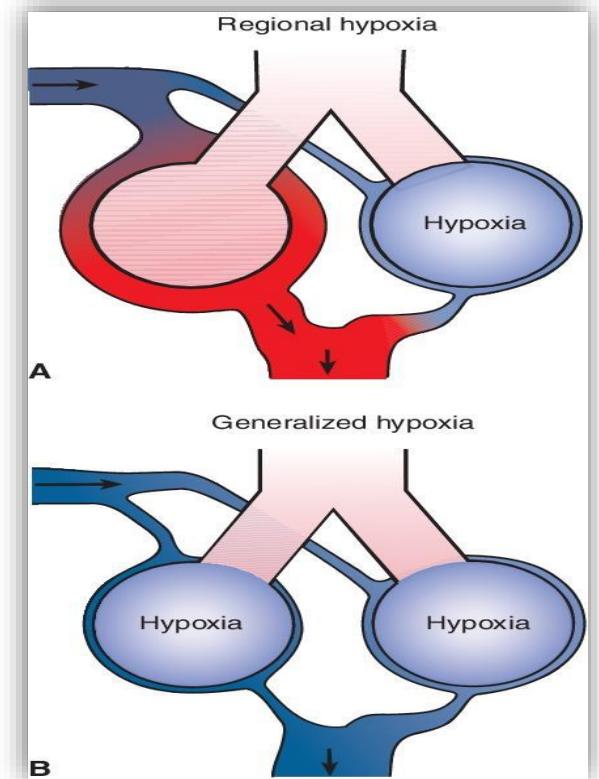
➡ Pulmonary Resistance = 7 mm Hg/83 ml/s = 0.08 peripheral resistance unit



Pulmonary vs. Systemic Circulation

-Response to Hypoxia

- Pulmonary vascular resistance can be increased by low oxygen tension.
- Both alveolar **hypoxia** and hypoxemia (low arterial PO_2) can directly stimulate pulmonary vascular smooth muscle cells and affect lung vascular resistance. The exact mechanism is not known
- This unique phenomenon is called **hypoxia-induced pulmonary vasoconstriction**
- In the **systemic circulation**, low oxygen tension causes **vasodilation**



Hypoxia-induced vasoconstriction can be regional affecting only part of the lung or generalized, affecting the entire lung.

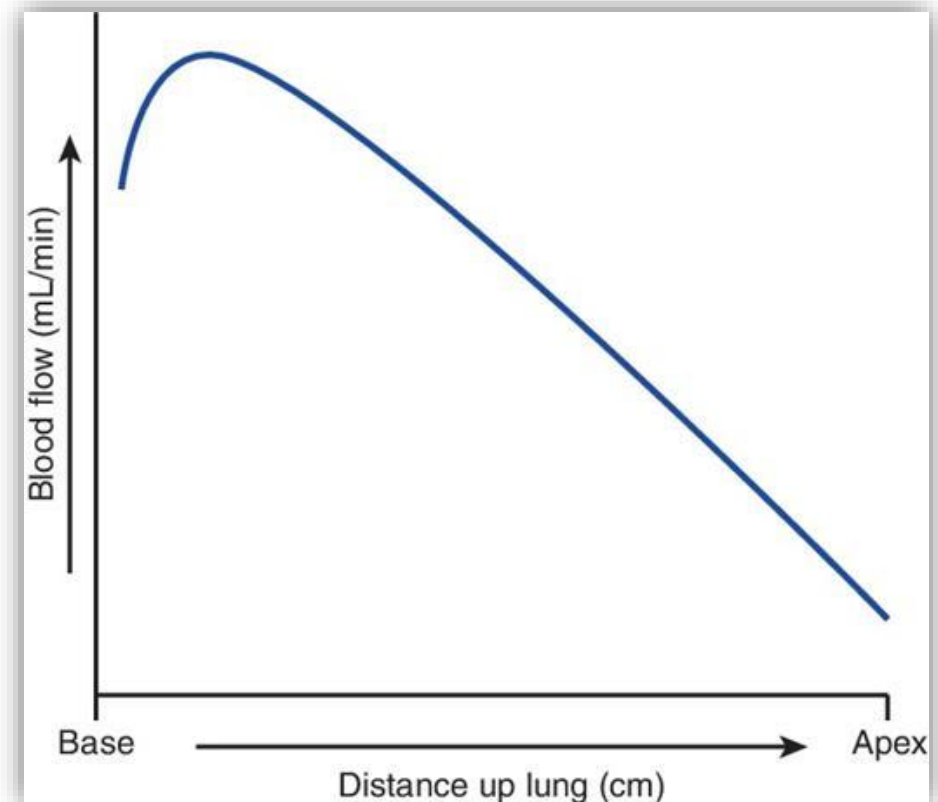
Summary: Objective 1

	Pulmonary Circulation	Systemic Circulation
Mean arterial pressure	Low	High
Resistance	Low	High
Compliance	High (less smooth muscle in arteriolar wall)	Low
Response to Hypoxia	Vasoconstriction	Vasodilation



Pulmonary Blood Flow and Lung Zones

- Gravity impacts pulmonary circulation by causing an uneven distribution of blood flow in the lungs
- In an upright person, gravity causes the blood flow to the base of the lung to be greater than the apex of the lung.
- The lung can be divided into zones to describe the gravitational effects of pulmonary capillary blood flow



Gravity causes uneven pulmonary blood flow in the upright individual.

The downward pull of gravity causes a lower blood pressure at the apex of the lungs. Consequently, pulmonary blood flow is low at the apex. Toward the base of the lungs, gravity has an added effect on blood pressure, causing an increase in blood flow. Hemodynamic features of the pulmonary circulation

Pulmonary Blood Flow and Lung Zones

Zone 1

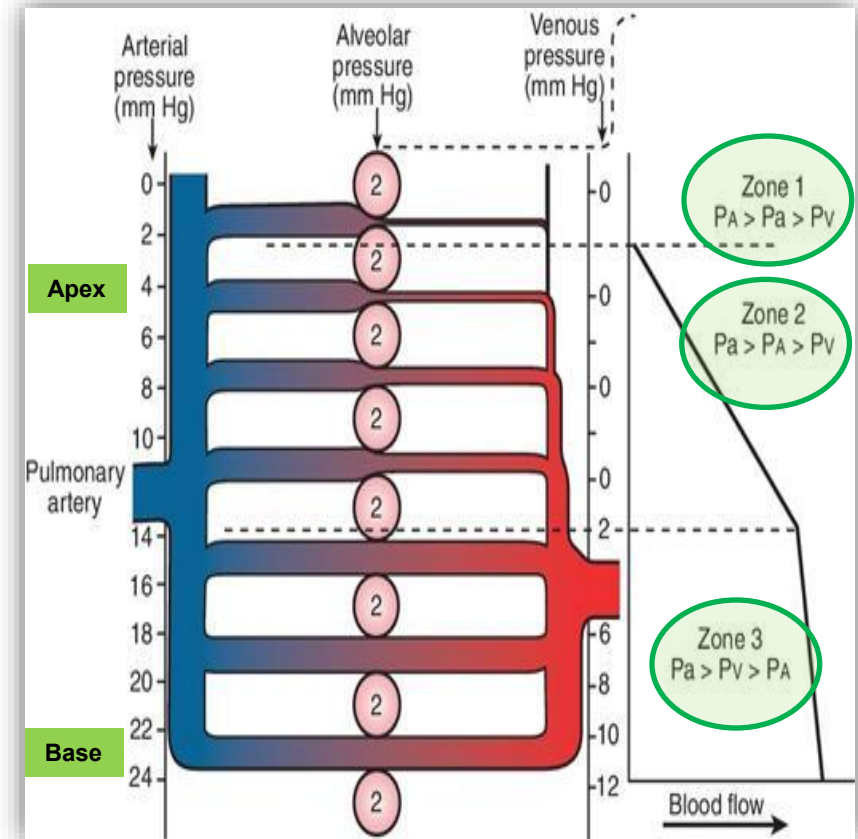
- Toward the apex of the lung
- Alveolar pressure exceeds arterial pressure
- There is no blood flow.

Zone 2

- Occurs in the middle of the lung
- Arterial pressure exceeds alveolar pressure,
- Blood flow depends on the difference between arterial and alveolar pressures.
- Blood flow is greater at the bottom than at the top of this zone.

Zone 3,

- Seen at the base of the lung
- Arterial and venous pressures exceed alveolar pressure, and blood flow depends on the normal arterial-venous pressure difference.
- Capillaries are significantly distended

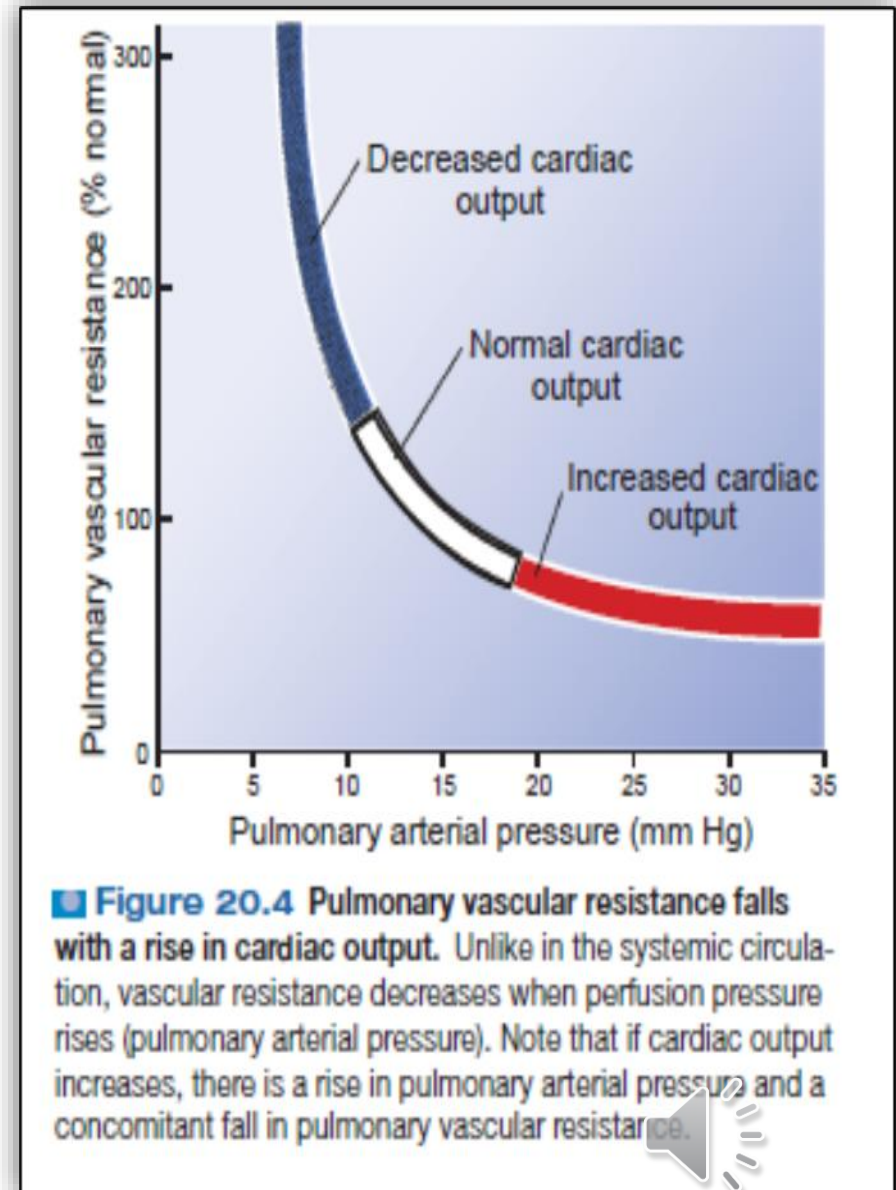


Lung zones are created due to gravitational effects.

The three lung zones in an upright person are dependent on the relationship between pulmonary arterial pressure (Pa), pulmonary venous pressure (Pv), and alveolar pressure (PA).

Pulmonary Recruitment and Distension

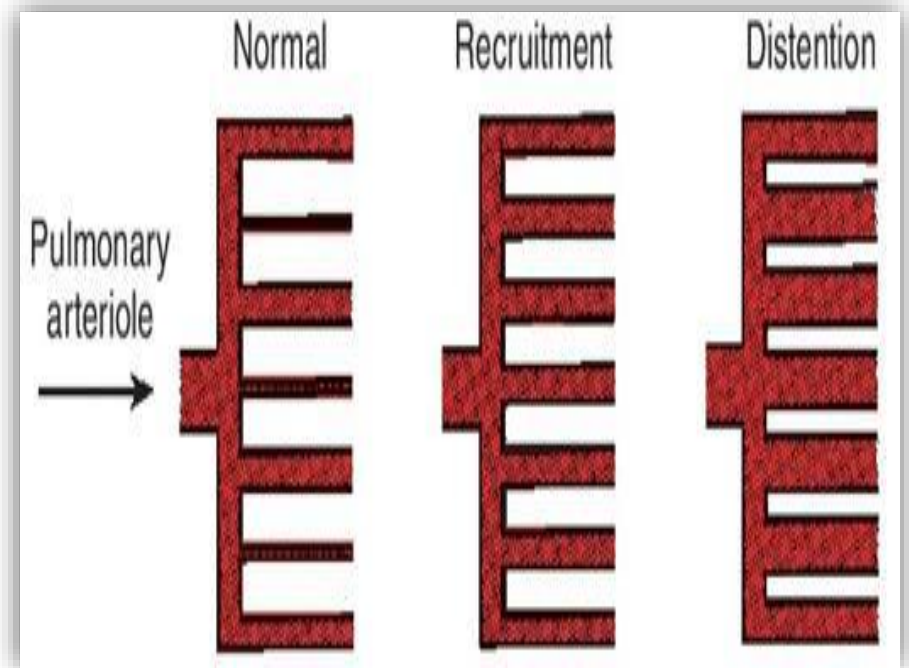
- An increase in cardiac output results in decreased pulmonary vascular resistance
- This is achieved by **recruitment** and **distension** of pulmonary capillaries
- This is beneficial because of:
 1. Increased capillary surface area, which enhances gas exchange
 2. Protection against high capillary pressures that can cause pulmonary edema



Explain how distention and recruitment of pulmonary vessels influences pulmonary vascular resistance with alterations in cardiac output or pulmonary arterial pressure. Explain in terms of distention and recruitment of pulmonary vessels. Identify the zones in which these two mechanisms apply.

Pulmonary Recruitment and Distension

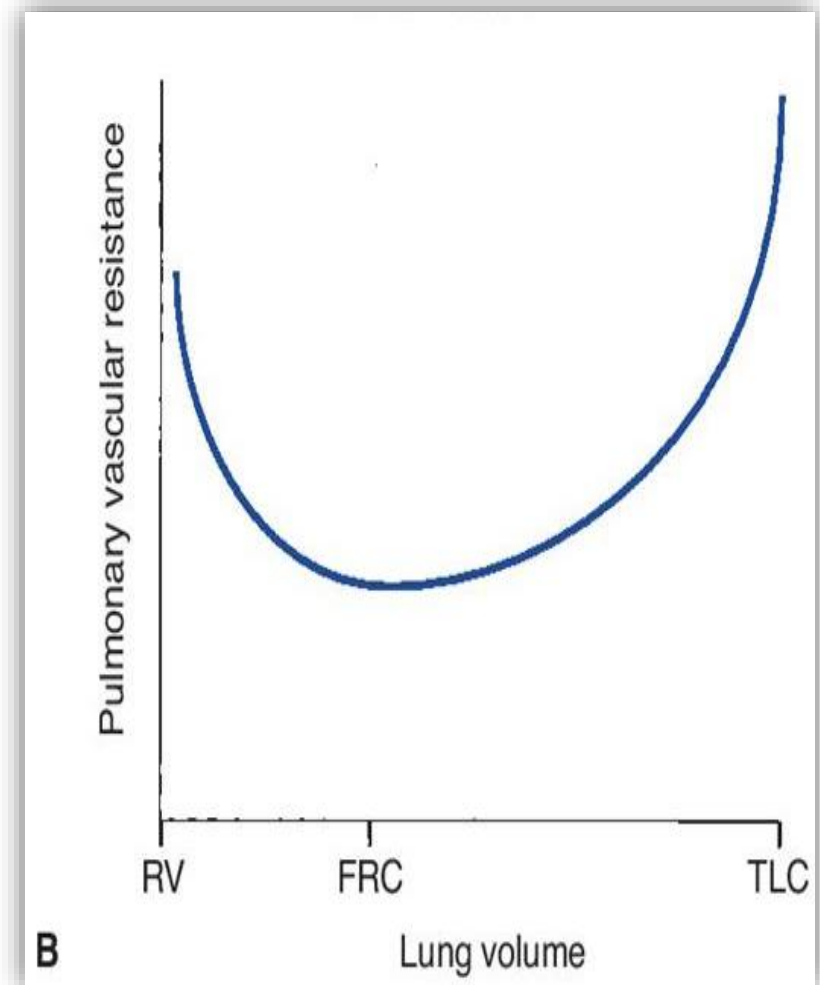
- Increased cardiac output results in increased pulmonary arterial pressure
- This increase in pressure causes the **apical capillaries** to open (**recruitment**)
- There is also widening of capillary segments due to high compliance of pulmonary vessels (**distension**)
- These changes cause a decrease in the pulmonary vascular resistance.



Under normal condition in an upright individual, not all capillaries are open or perfused at the apex. Capillary recruitment (the opening up of previously closed vessels) results in the perfusion of an increased number of vessels with a concomitant decrease in resistance. Capillary distention (an increase in the caliber of vessels) resulting from high vessel compliance also contributes to the low resistance.

Pulmonary Vascular Resistance and Lung Volumes

- Lung volumes significantly affect pulmonary vascular resistance (PVR).
- This is due to the pulmonary architecture. The small pulmonary vessels (capillaries, arterioles and venules) are connected to the lung tissue.
- The vessels can therefore be easily distended or collapsed based on the pressure of the surrounding tissue.

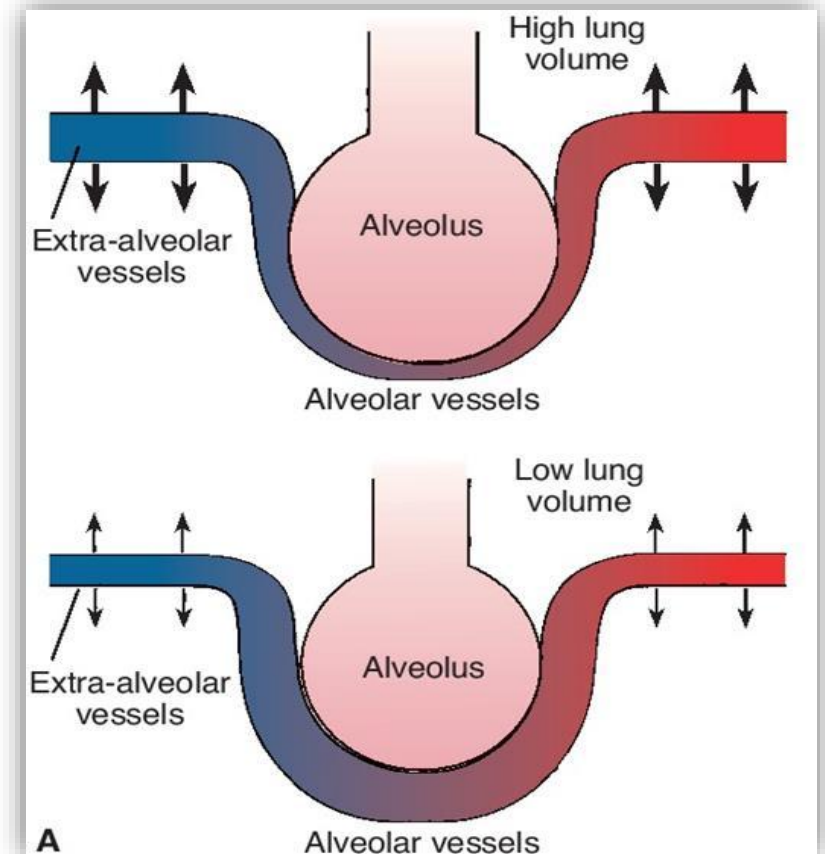


Pulmonary vascular resistance increases at high and low lung volumes.

The effect of lung volumes on pulmonary vascular resistance follows a U-shaped curve, with resistance being the lowest at functional residual capacity (FRC). RV, residual volume; TLC, total lung capacity.

Pulmonary Vascular Resistance and Lung Volumes

- The pulmonary vessels can be classified as
 1. Extra-Alveolar (pulmonary artery and vein)
 2. Alveolar vessels (pulmonary arterioles, capillaries and venules)
- Alveolar vessels' diameter can be modified by alveolar pressure
- Extra-alveolar vessels' diameter can be modified by pleural pressure
- Pulmonary vascular resistance = *resistance alveolar vessels + resistance in the extra-Alveolar vessels.*



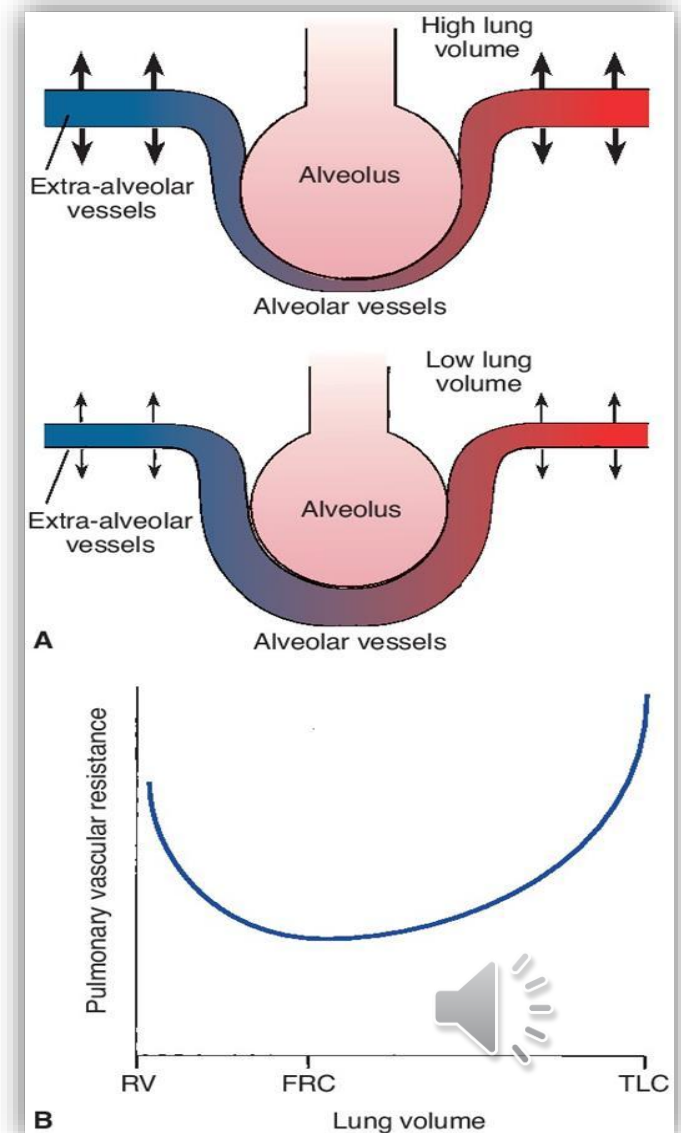
Pulmonary vascular resistance is influenced by changes in lung volumes.

At high lung volumes, extra-alveolar vessels are distended because of the lower pleural pressure. However, alveolar vessels are compressed, causing a rise in pulmonary vascular resistance. At low lung volumes, alveolar vessels are distended, but the extra-alveolar vessels are compressed from the rise in pleural pressure, which results in a rise in pulmonary vascular resistance.

Pulmonary Vascular Resistance and Lung Volumes

Lung Volume	Extra-Alveolar Vessels	Alveolar Vessels
High Lung Volumes (TLC) Net result = ↑ PVR	- More negative pleural pressure - Increased transmural pressure in the extra-alveolar vessels - Vessels distend	- Alveolar diameter increases - Decrease transmural pressure in alveolar vessels - Vessels are compressed causing increased PVR
Low Lung Volume (RV) Net result = ↑ PVR	- More positive pleural pressure - compression of extra-alveolar vessels - Increased PVR	- Alveolar diameter is decreased - Increased transmural pressure - Vessels distend

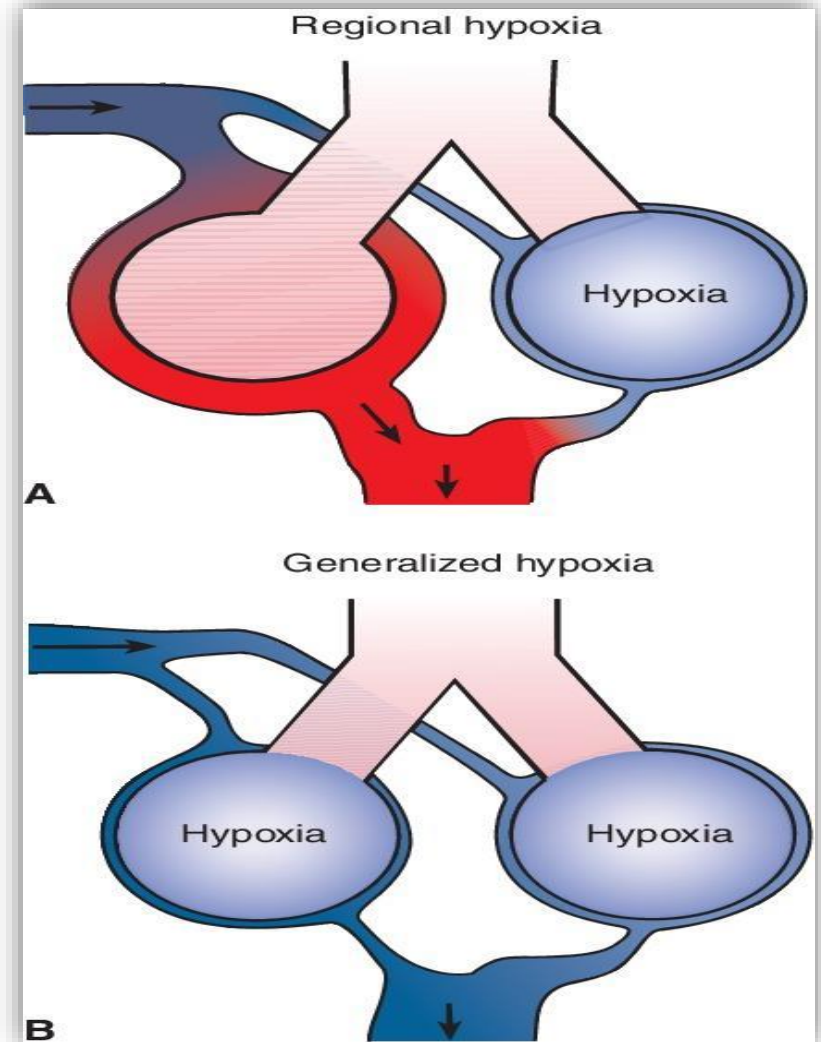
- Pulmonary vascular resistance is lowest at functional residual capacity (FRC) and is increased at higher and lower lung volumes.
- Pulmonary vascular resistance = *resistance alveolar vessels + resistance in the extra-Alveolar vessels.*



Pulmonary Blood Flow and Hypoxic Vasoconstriction

- Alveolar hypoxia causes pulmonary vasoconstriction
- The two types of alveolar hypoxia are **regional hypoxia** and **generalized hypoxia**
- Each type of alveolar hypoxia has a unique effect on the PVR

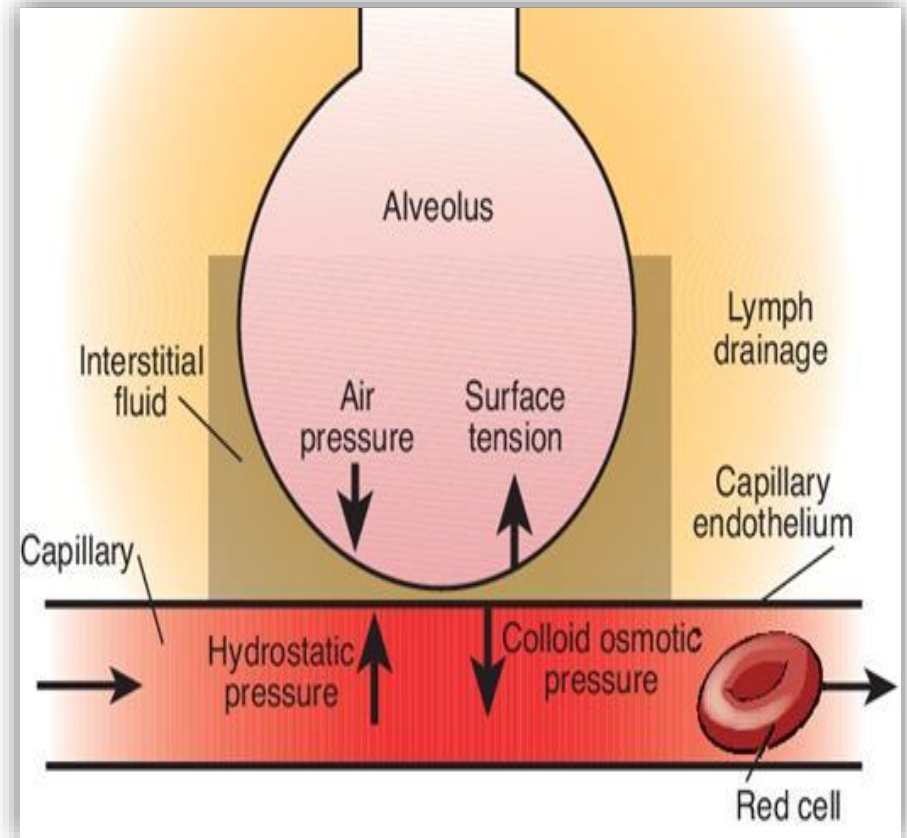
Regional Hypoxia	Generalized Hypoxia
- Localized vasoconstriction diverts blood from poorly ventilated region - Commonly caused by bronchial obstruction - little effect on PVR - blood flow is restored after the hypoxia resolves	- Vasoconstriction in both lungs - Commonly due to diseases such as asthma and emphysema and, high altitude - Significant rise in pulmonary artery pressure and PVR



With **regional hypoxia**, precapillary constriction diverts blood flow from poorly ventilated regions, with little change in pulmonary arterial pressure. In **generalized hypoxia**, which can occur with high altitude or with certain lung diseases, precapillary constriction occurs throughout the lungs, and there is a marked increase in pulmonary arterial pressure.

Pulmonary Edema

- Abnormal accumulation of fluid in lung interstitium and alveoli
- Edema results from either decreased oncotic pressure, increased hydrostatic pressure (starling forces), or increased capillary permeability
- In the lung the alveolar surface tension augments filtration and alveolar pressure opposes filtration.
- **Under normal condition:**
Small amount of fluid getting into the interstitium. This fluid is drained by lymphatics.



Fluid movement in and out of capillaries depends on the net difference between hydrostatic and colloidal osmotic pressures. In the lung, two additional factors (alveolar surface tension and pressure) are involved in fluid exchange. Alveolar surface tension enhances filtration, whereas alveolar pressure opposes filtration. The relatively low pulmonary capillary hydrostatic pressure helps keep the alveoli "dry" and prevents pulmonary edema.



Pulmonary Edema

- Oncotic pressure and alveolar pressure favors reabsorption
- Filtration is promoted primarily by the alveolar surface tension because the pulmonary hydrostatic pressure is lower than the oncotic pressure
- Pulmonary edema results when capillary filtration exceeds fluid removal
- Pulmonary edema is classified as cardiogenic (due to heart dysfunction) or noncardiogenic (due to lung injury).
- Cardiogenic pulmonary edema is most commonly due to increased pulmonary capillary hydrostatic pressure
- Noncardiogenic pulmonary edema arises due to direct or indirect insult to the membrane permeability of the pulmonary capillary

Etiology	Examples of Disease States
Increased Pulmonary capillary Hydrostatic Pressure	Mitral stenosis, Conditions causing left ventricular failure with elevated left atrial pressure, High altitude pulmonary edema.
Reduced Oncotic Pressure	Hypoalbuminemia, Malnutrition, Cirrhosis of liver, Nephrotic syndrome.
Reduced Lymphatic Drainage	Tumors in the thorax obstructing lymphatics.
Loss of lung surfactants and increased surface Tension	Acute respiratory distress syndrome in both infants and adults.
Capillary Permeability increased	Inflammation due to inhalation of lung irritants, Acute respiratory distress syndrome.

Review Questions

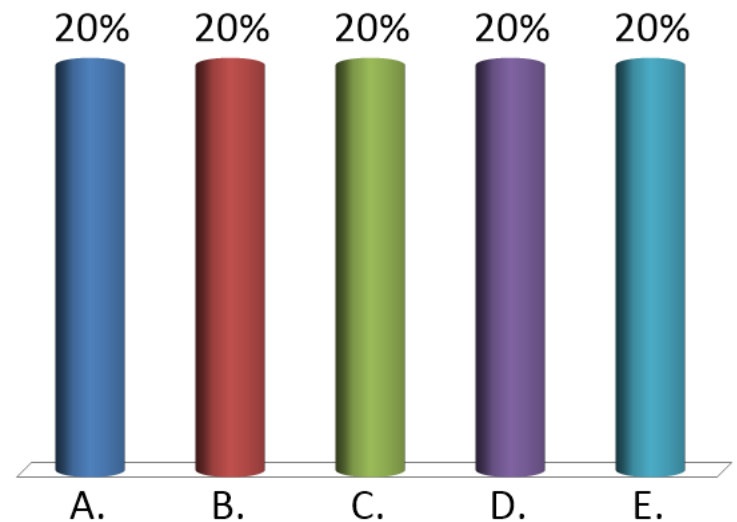
Pulmonary Edema: Consequences

- Excess fluid in the lung interstitium increases thickness of the respiratory membrane and hinders gas exchange, causing hypoxemia. Hypercapnia may not occur as the CO₂ is a better diffusible gas than oxygen.
- Flooding of small airways obstruct airflow and increase airway resistance.
- Decreases lung compliance due to interstitial swelling and the increase in alveolar surface tension.
- Increases the work of breathing.
- Management is aimed at reducing pulmonary capillary hydrostatic pressure, improving oncotic pressure and preventing hypervolemia.



A 76-year-old man comes to the physician because of shortness of breath and swollen legs for 3 weeks. His pulse 98/min, respirations are 23/min and blood pressure is 150/100 mm Hg. Physical examination show decreased breath sounds at the bases bilaterally and 3+ pitting edema. Investigations lead to a diagnosis of pulmonary edema secondary to liver failure. Which of the following changes is the most likely cause of the pulmonary edema in this patient?

- A. Increased Capillary Permeability
- B. Increased Hydrostatic Pressure
- C. Increased Surface Tension
- D. Reduced Oncotic Pressure
- E. Reduced Lymphatic Drainage



A 61-year-old man comes to the emergency room because of a 1-week history of shortness of breath and productive cough. His blood pressure is 138/82 mm Hg, respirations are 16/min, and his pulse is 98/min. Physical examination shows no signs of respiratory distress and auscultation shows no abnormality. Which of the following changes in alveolar pressure and pulmonary vascular resistance are most likely to occur when this patient inspires to maximum lung volume?

	Alveolar Pressure	Pulmonary vascular resistance
A.	↑	↑
B.	↑	↓
C.	↓	↓
D.	↓	↑
E.	↑	↔
F.	↔	↓

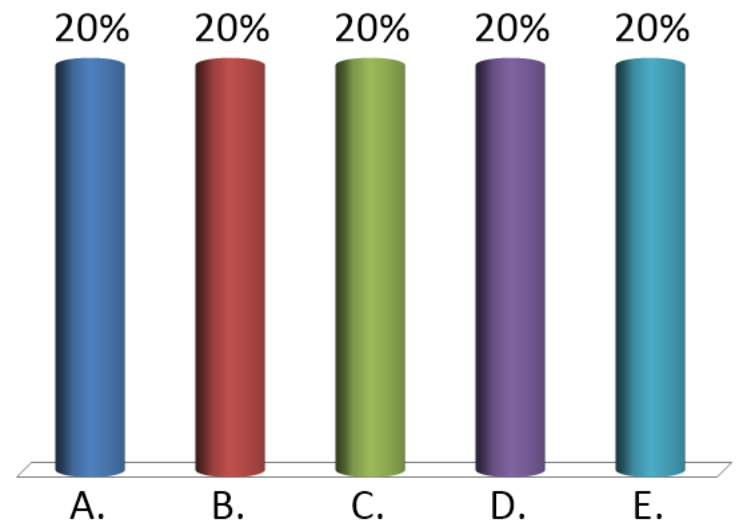
A 21-year-old woman traveled from sea level a remote mountain community at very high altitude in one day. Which of the following findings is most likely to occur in this traveler on arrival at this location?

- A. Decreased respiration
- B. Generalized hypoxic vasoconstriction in the lung
- C. Increased arterial O_2 (PaO_2)
- D. Regional vasodilation in the lung
- E. Decreased work of breathing

Answers

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